

What the bounded calibration refinement achieved

Same model, targets, weights, and external restrictions

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The result for discussion

The best verified candidate has loss **23.153445**, compared with **30.482967** for the retained production calibration: a **24.04% reduction** in the same twelve-row objective. Two fresh executions reproduce every fit row, parameter, preference normalization, and saved numerical history entry exactly. This is a bounded local improvement, with weak parameter identification still unresolved. It is recorded for review; production has not been silently replaced.

The first-birth housing response moves from **0.4364** to **0.4660 rooms**, against **0.7202** targeted. Mean occupied rooms move from **6.3173** to **6.3286**, against **5.7800** targeted. The first-birth response explains **64.7%** of its target. A lower total objective must be assessed alongside this remaining gap.

The largest gain comes from the rooms contrast between families with three or more children and families with one or two: its model value changes from **0.4034** to **0.3792 rooms**, against **0.3677** targeted. Mean occupied rooms remain too high and become slightly higher at this candidate.

Evaluation stage	Completed cases	Best loss in stage
Inherited-candidate exact repeats	2	29.385884
Small coordinate moves	23	28.587975
Joint parameter moves	12	23.153445
Final exact repeats	2	23.153445

All **39** completed evaluations retain the original source and target fingerprints, twelve moments, eleven free parameters, the 120-point wealth grid, dated housing-supply elasticity 0.63, and tenure-choice dispersion 0.005. They reconstruct the old state and full 2007-2023 path. Every case has a terminal checkpoint, complete target and parameter tables, and the unchanged seventeen-figure packet. No policy path or long-run response was recalculated in this exercise.

Where the fit changes

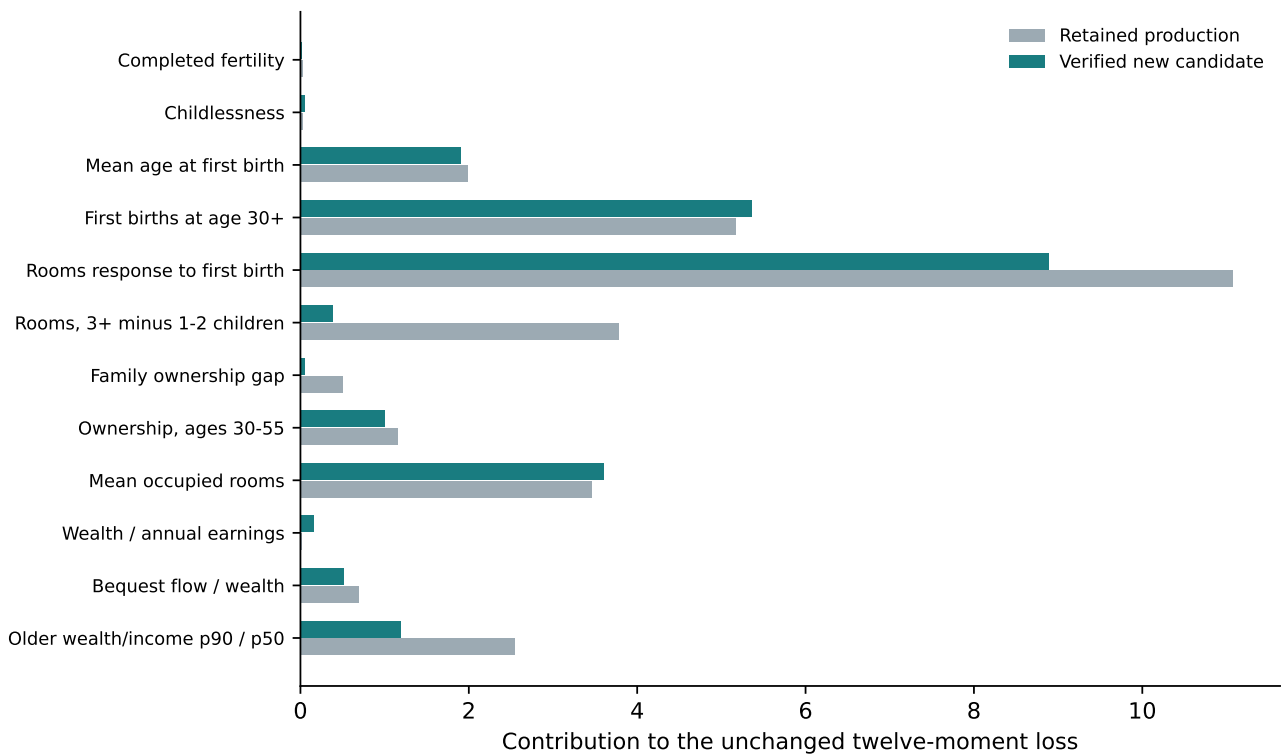


Figure 1: Supplemental comparison of every objective contribution. Both series use identical target values and weights.

The first round tests small separate parameter moves. The second combines successful moves and tests a child-space rebalancing: increase the first-child intercept while reducing the per-child floor. With $\bar{h}(m) = \bar{h}_J + m\bar{h}_C$ for children at home $m > 0$, the proposed changes $d\bar{h}_J = d$ and $d\bar{h}_C = -d/3$ preserve the three-child floor and raise the first-child floor by $2d/3$. Actual occupied rooms still depend on prices, saving, tenure choice, and fertility selection; the floor identity is not an identity for observed housing use.

All ranking uses the complete unchanged objective. No target was removed, reweighted, or added. The earlier ridge planner's rejection is retained; these direct candidate evaluations do not invert its unstable Jacobian or establish precise estimates of the weakly informed bequest parameter.

Complete target fit: verified candidate

Gaps equal model minus target. Contributions are weight times squared gap. The weights retain their existing empirical or synthetic interpretations; this loss is not a specification-test statistic.

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918000	1.921698	+0.003698	1425.74	0.0195
Childlessness	0.188000	0.189859	+0.001859	17180.7	0.0594
Mean age at first birth	26.044627	26.251703	+0.207076	44.4444	1.9058
First births at age 30+	0.260327	0.237163	-0.023165	10000	5.3660
Rooms response to first birth	0.720246	0.466044	-0.254202	137.565	8.8893
Rooms, 3+ minus 1-2 children	0.367700	0.379198	+0.011499	2958.51	0.3912
Family ownership gap	0.167662	0.169578	+0.001917	14229.6	0.0523
Ownership, ages 30-55	0.575472	0.546618	-0.028854	1207.85	1.0056
Mean occupied rooms	5.779970	6.328594	+0.548624	11.9732	3.6038
Wealth / annual earnings	6.873100	7.031196	+0.158096	6.28767	0.1572
Bequest flow / wealth	0.008800	0.008485	-0.000315	5.16529e+06	0.5139
Older wealth/income p90 / p50	3.448111	3.303595	-0.144516	56.9598	1.1896

Complete target fit: retained production

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918000	1.922907	+0.004907	1425.74	0.0343
Childlessness	0.188000	0.189407	+0.001407	17180.7	0.0340
Mean age at first birth	26.044627	26.256032	+0.211404	44.4444	1.9863
First births at age 30+	0.260327	0.237579	-0.022748	10000	5.1748
Rooms response to first birth	0.720246	0.436418	-0.283828	137.565	11.0821
Rooms, 3+ minus 1-2 children	0.367700	0.403441	+0.035741	2958.51	3.7793
Family ownership gap	0.167662	0.161699	-0.005963	14229.6	0.5060
Ownership, ages 30-55	0.575472	0.544488	-0.030984	1207.85	1.1596
Mean occupied rooms	5.779970	6.317291	+0.537321	11.9732	3.4568
Wealth / annual earnings	6.873100	6.932652	+0.059552	6.28767	0.0223
Bequest flow / wealth	0.008800	0.008433	-0.000367	5.16529e+06	0.6971
Older wealth/income p90 / p50	3.448111	3.236511	-0.211600	56.9598	2.5503

Complete parameter comparison

Parameter	Production	New candidate	Bounds / restriction	Near bound?
Annual patience β	0.995117	0.995739	[0.94, 0.9995]	No
First-birth dispersion κ_1	2.168173	2.168173	[0.02, 50]	No
Later-birth dispersion κ_C	1.736471	1.736471	[0.02, 50]	No
Owner-service premium χ	1.043472	1.043472	[0.1, 5]	No
Supply intercept H_0	14.562959	14.562959	[0.2, 80]	No
Bequest strength θ_0	0.528428	0.549189	[0, 8]	No
Bequest shifter θ_1	0.107249	0.107249	[0.02, 16]	Raw scale only
Per-child room floor	0.282210	0.248877	[0.1, 1.8]	No
First-birth fixed cost	4.559138	4.559138	[0, 8]	No
First-child room intercept	0.364931	0.464931	[0, 0.5]	No
2007-2023 child-value change	-0.328714	-0.328714	[-1.5, 0.2]	No
Tenure dispersion κ	0.005000	0.005000	Externally fixed	Not estimated
Dated supply elasticity η	0.630000	0.630000	Externally fixed	Not estimated

The old fertility-preference intercept is re-normalized to replacement for every parameter vector. New candidate values are $\psi_{2007} = 0.28947879$ and $\psi_{2023} = -0.03923512$. The stored near-bound flag uses raw parameter units. For the bequest shifter, interpret it alongside its position in the logarithmic search interval; a raw-scale flag alone does not establish a binding search constraint.

The supply restriction 0.63 applies to dated markets. The inherited initialization retains elasticity 1.75 before the dated supply intercept is normalized. Both values and that order are unchanged across this comparison; whether this is the intended economic contract remains a discussion item in the separate code audit.

The complete smaller-step panel has algebraic Jacobian rank **11 of 11** and relative rank **11 of 11** at the declared 10^{-3} threshold. These are local, step-dependent diagnostics, not standard errors or proof of global identification. The saved receipt includes every singular value, weak-direction loading, and forward/backward derivative comparison. The weakest direction has **98.4%** of its squared loading on the bequest shifter. The supply-intercept derivative has forward/backward cosine **-0.369**; opposite directional responses make a smooth local inverse unreliable here.

Numerical checks and their limits

All completed cases pass the unchanged production acceptance checks. Those gates do not certify every policy function. The separate adjacent-wealth screen flags occupied value decreases in 2 evaluations; the largest exposed pre-choice mass share is **2.08e-07**. These flags are retained and are consistent with the already documented need to check the local saving optimizer; this search does not establish their cause through a new global-optimizer solve. The selected candidate has 0 flagged occupied value decreases. Its reporting-floor budget-excess share is **1.76e-09**. Probability arrays are finite and remain inside $[0, 1]$. The seventeen standard graphs follow this note unchanged.

The standard age-30 first-birth panel also shows a probability spike at 1 negative-wealth renter state. Its settled-childless pre-choice mass is 0. This is a check of current exposure, not a proof of optimal policies outside occupied states. The original graphs and their dense income-state labels are preserved.

The concurrent source-code audit also reproduces an incomplete policy-storage interface: later-birth probabilities reside in a mutable parameter field instead of the saved calendar policy. Reuse after an intervening solve can mix policies. No affected result has been demonstrated in this historical calibration: the nonzero preference drift clears warm starts at each later date. The interface requires a separate repair and targeted replay before relying on vulnerable policy-reuse paths. Exact reproduction alone would not detect this defect.

What remains to discuss

The first-birth rooms contrast remains a provisional empirical-to-model mapping. The young-ownership diagnostic is still excluded from estimation; model and empirical age windows need a common contract, including existing prime-age and old-wealth statistics. The overnight evidence on near-universal late-life ownership and sensitivity to rental opportunities remains relevant. A better in-sample objective does not independently resolve those issues.

Supply and credit results in the earlier morning audit belong to the retained production calibration. This refinement supplies no revised policy magnitudes, welfare ranking, or resolution of the household-entry transition.

The complete per-case fit and parameter tables are in each stage's `report/all_target_fits.csv` and `report/all_parameters.csv` under `output/model/e5f_morning_refinement_20260905a/`. Its README provides stage plans, execution receipts, checkpoints, standard plots, and reproduction commands. The selected case is **round2/task_009/summary.json**.

Standard solution diagnostics

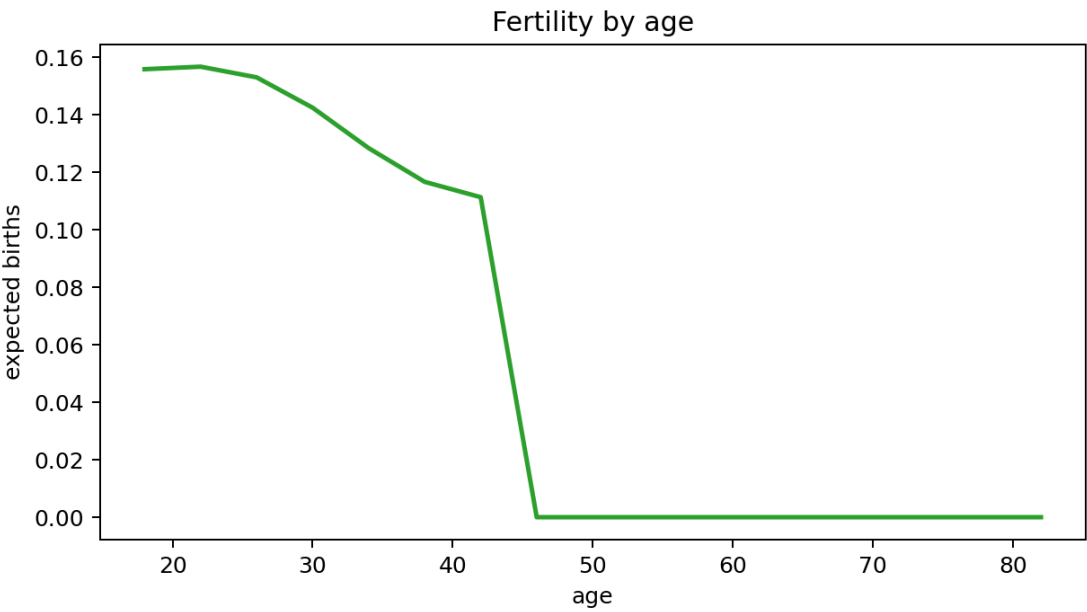


Figure 2: Fertility by age.

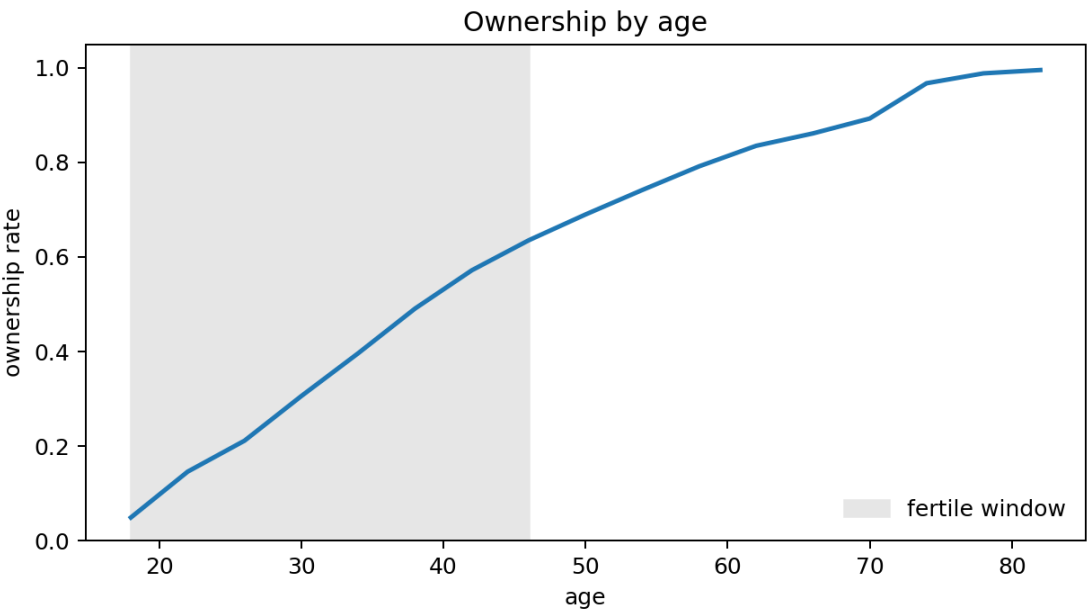


Figure 3: Ownership by age.

Standard solution diagnostics

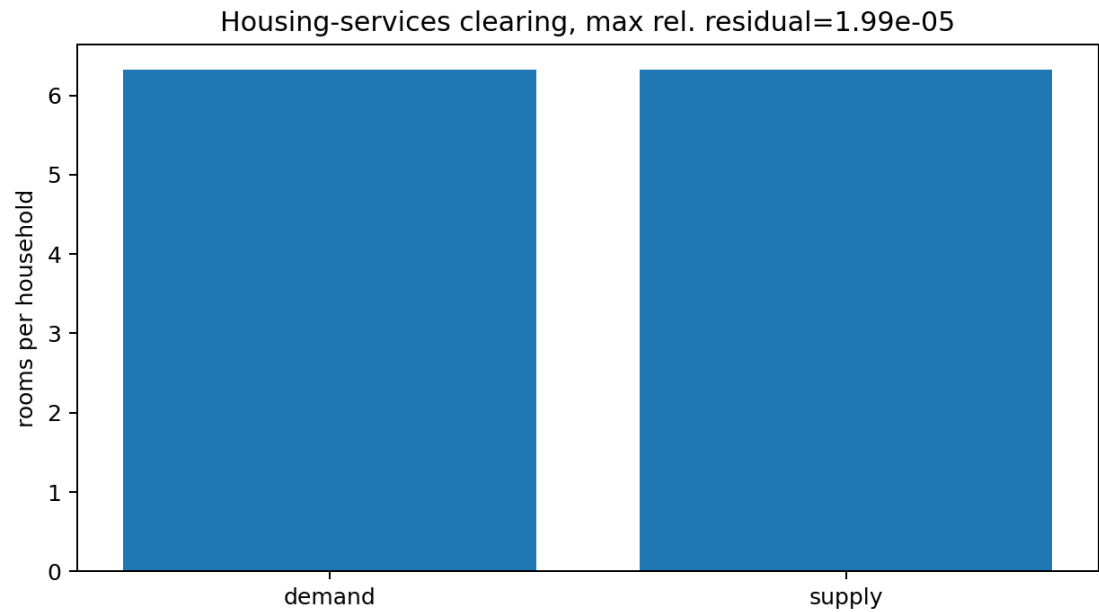


Figure 4: Housing market.



Figure 5: Housing prices.

Standard solution diagnostics

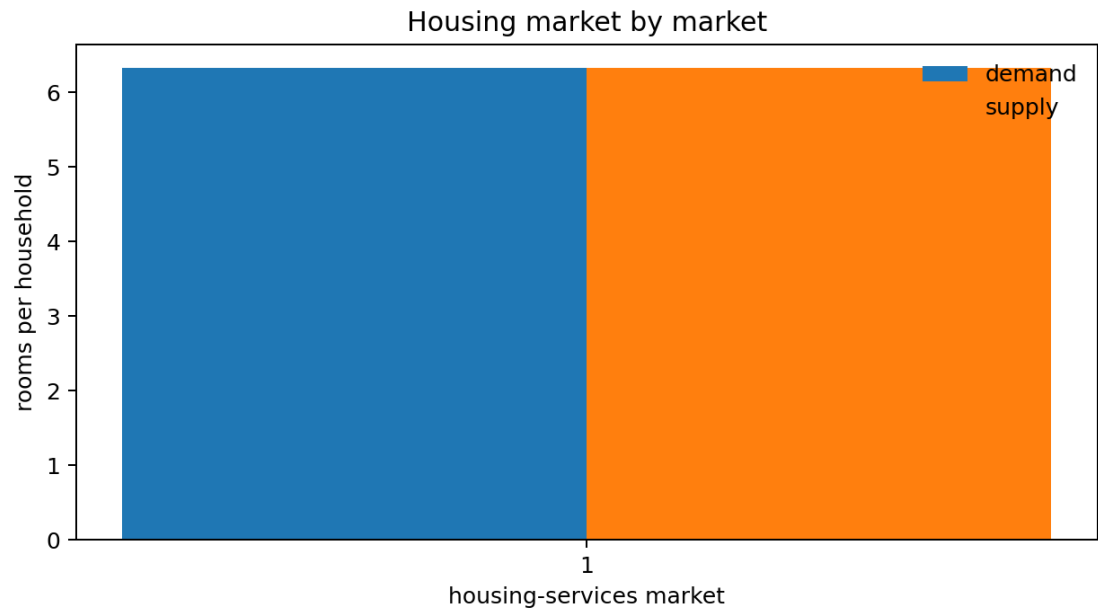


Figure 6: Market clearing by market.

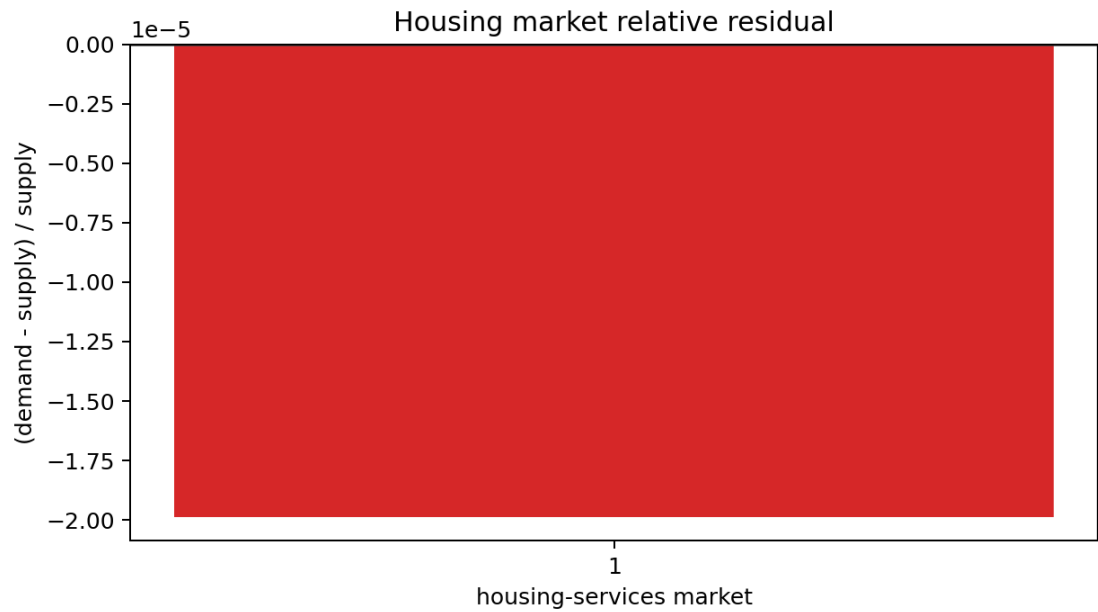


Figure 7: Market clearing residuals.

Standard solution diagnostics

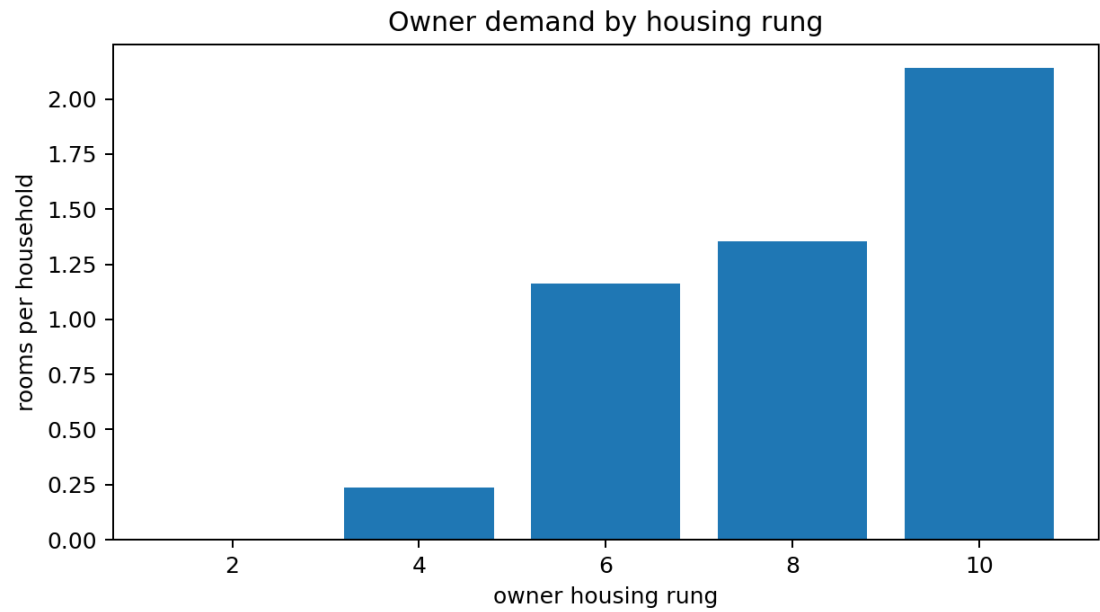


Figure 8: Owner rungs.

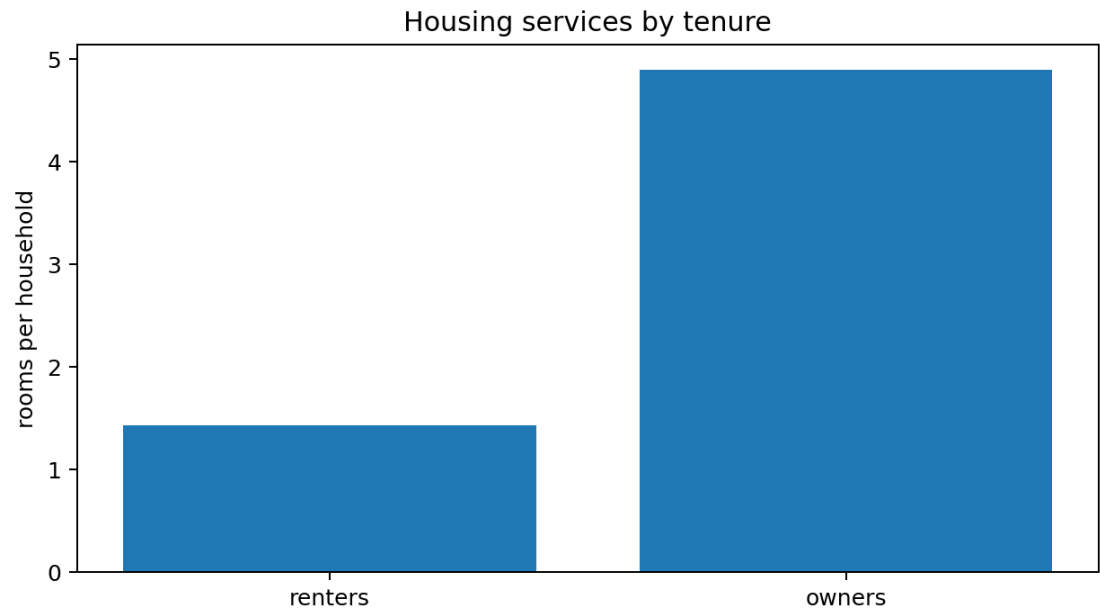


Figure 9: Tenure services.

Standard solution diagnostics

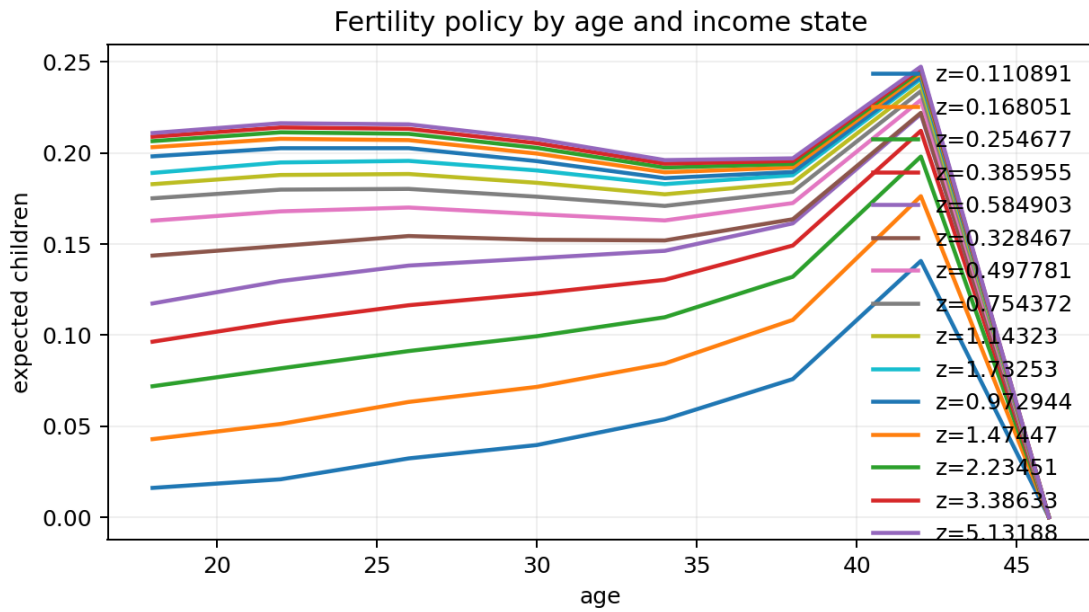


Figure 10: Fertility policy by age income state.

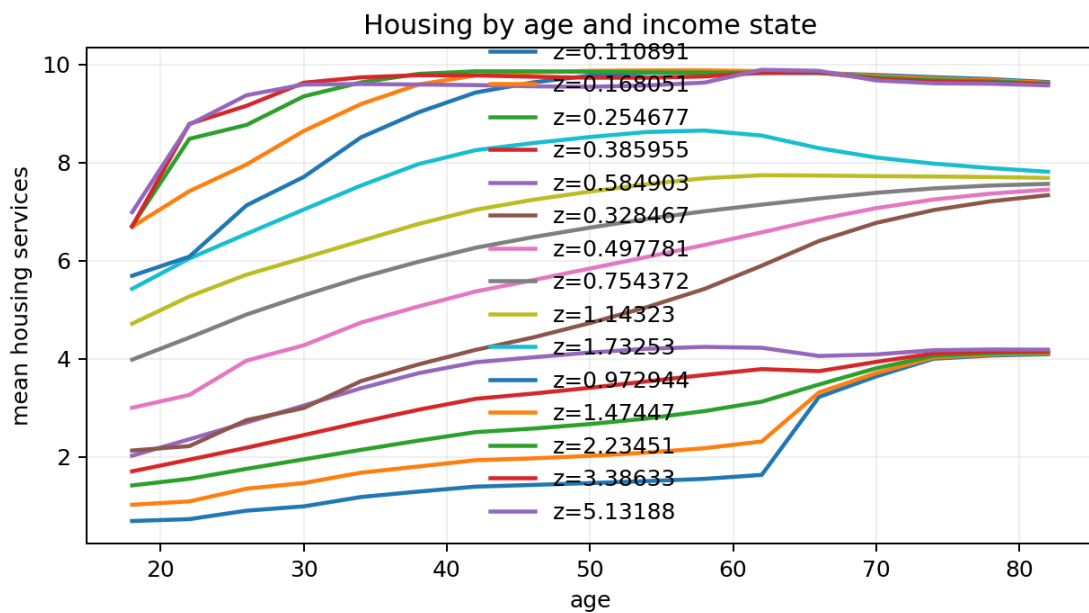


Figure 11: Housing by age income state.

Standard solution diagnostics

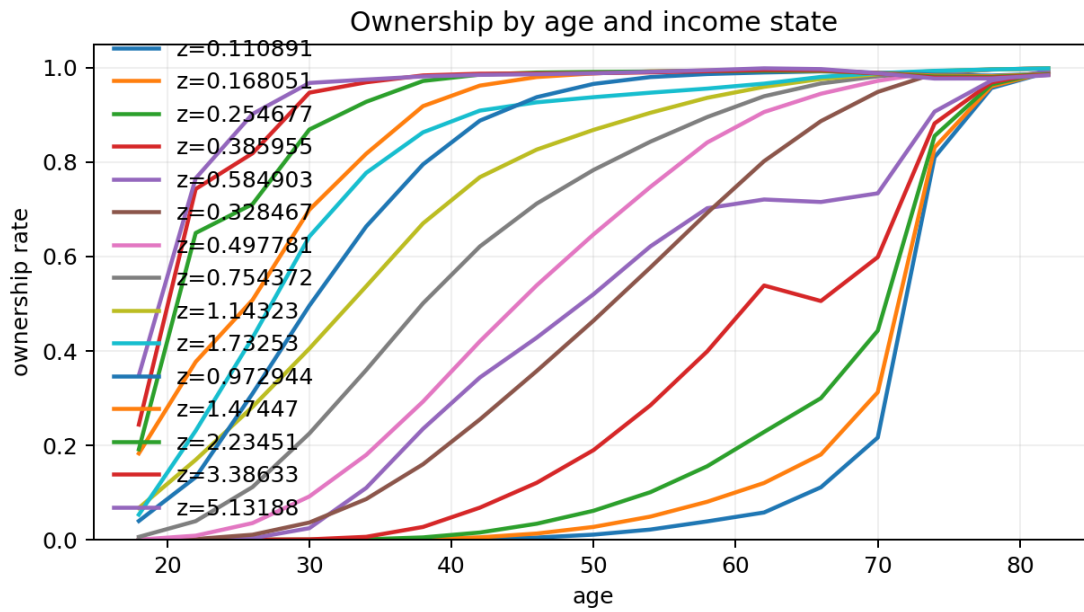


Figure 12: Ownership by age income state.

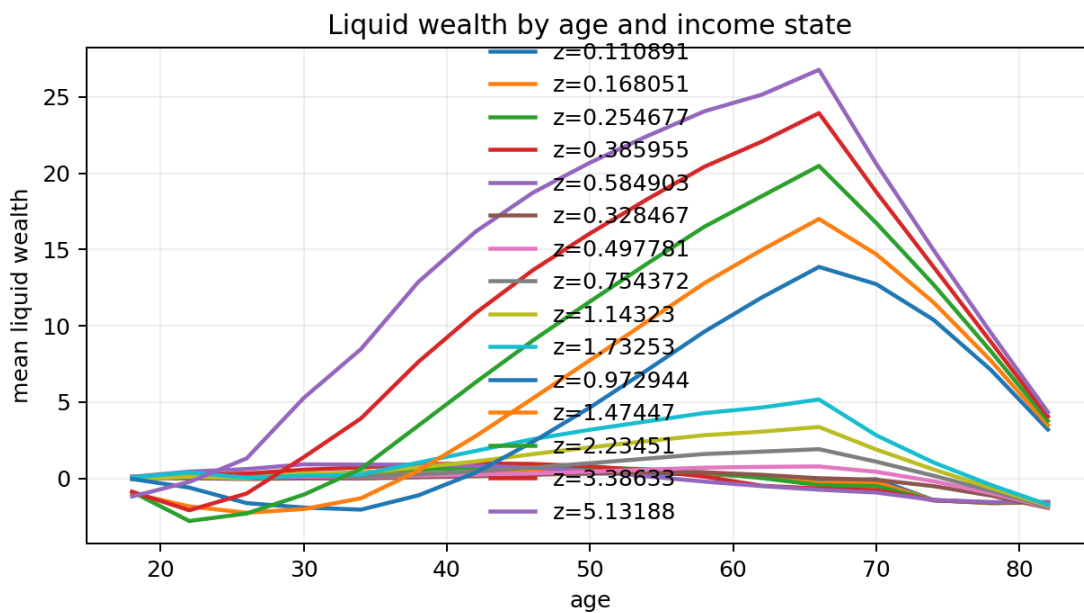


Figure 13: Liquid wealth by age income state.

Standard solution diagnostics

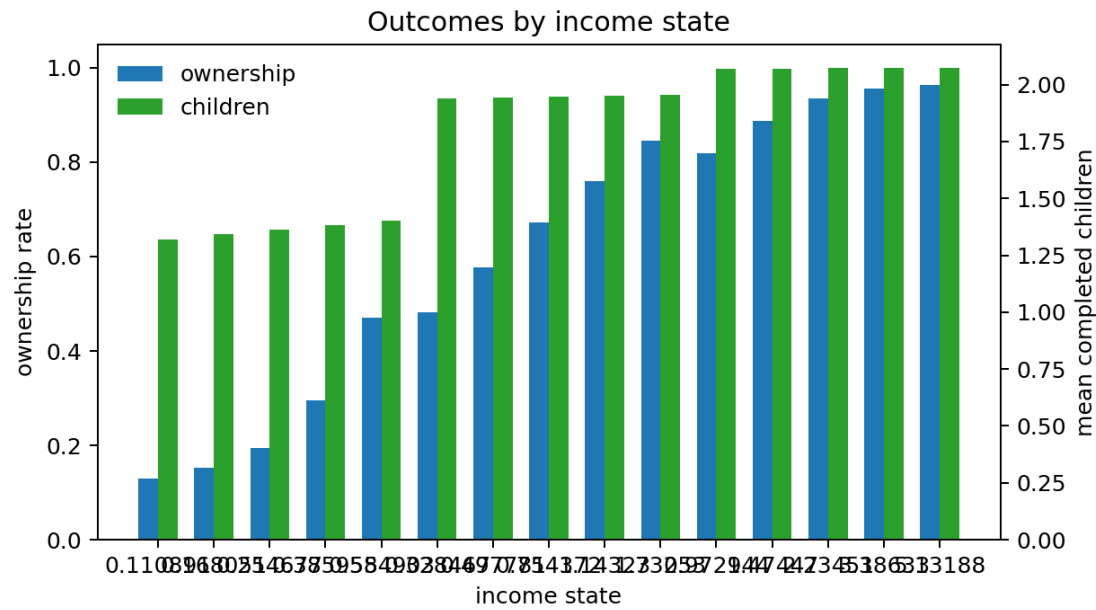


Figure 14: Income state outcomes.

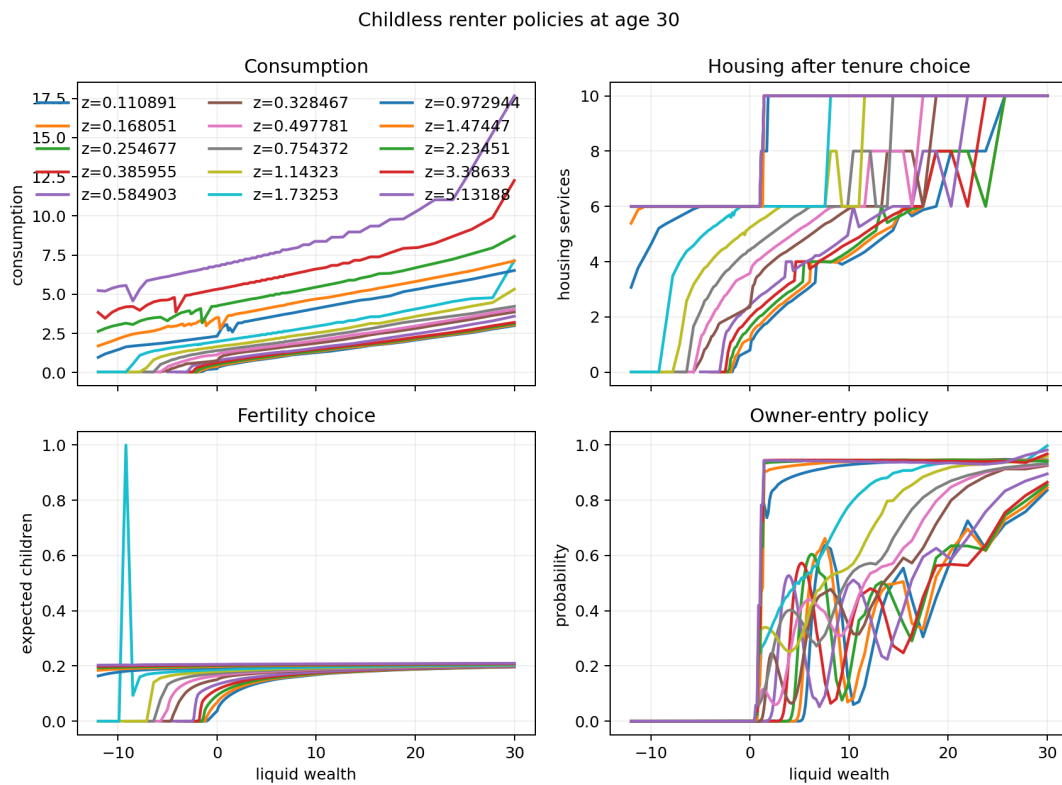


Figure 15: Policy childless renter age30.

Standard solution diagnostics

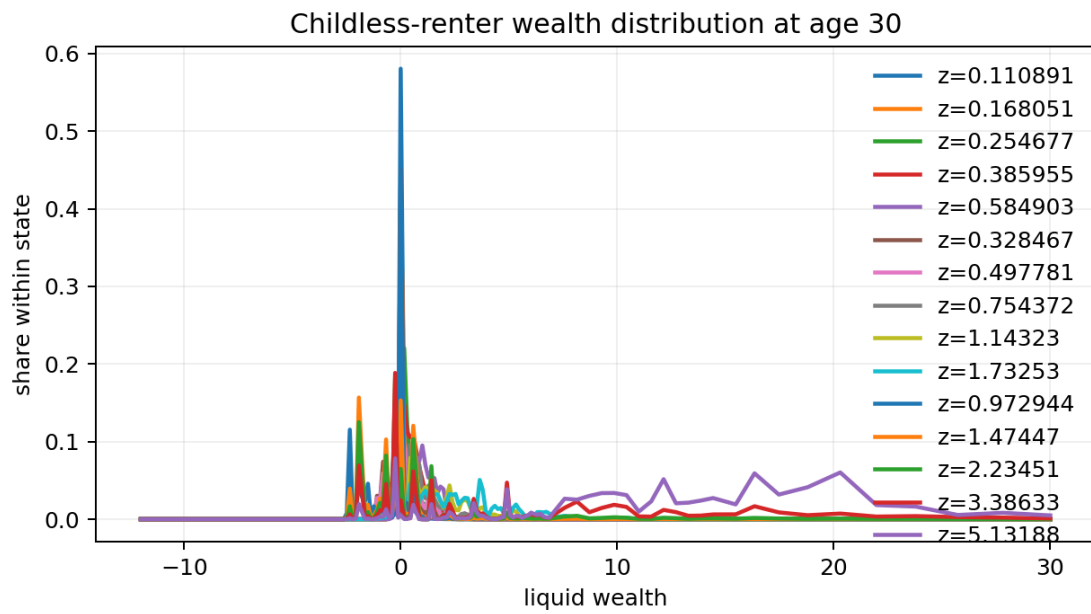


Figure 16: Wealth dist childless renter age30.

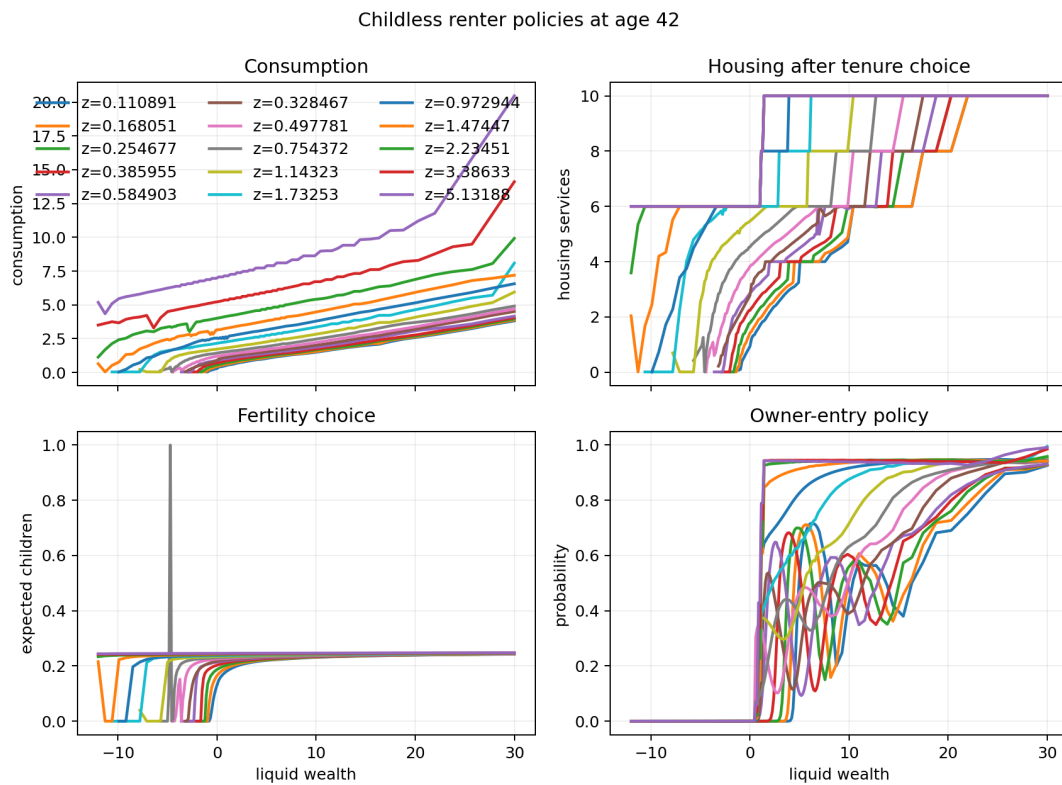


Figure 17: Policy childless renter age42.

Standard solution diagnostics

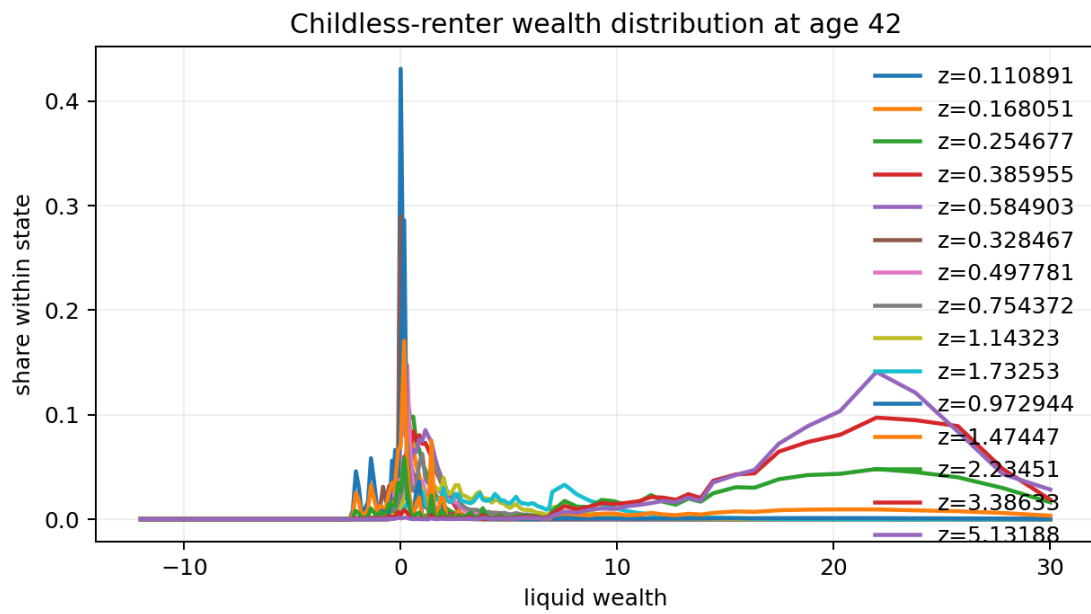


Figure 18: Wealth dist childless renter age42.